

BINARY TREE TRAVERSAL

Visualizers & Practice Tools

A curated guide to seeing traversal paths in action, and generating unlimited practice challenges.



Interactive Visualizers

See exactly how traversals move through nodes, in real time

1

VisuAlgo — Binary Search Tree

Create custom trees, choose a traversal algorithm (Pre / In / Post / Level-order), and watch an animated highlight step through each node.

2

Binary Tree Visualizer and Converter

Type an array or a list of nodes to instantly see the drawn tree alongside its calculated traversal outputs.

3

Data Structure Visualizations (USF)

A highly interactive tool for inserting nodes one by one and manually tracing each step of a traversal's execution.

1. VisuAlgo – BST & AVL Tree Visualizer

VISUALGO

DATA STRUCTURES

Array

Linked List

Stack

Queue

Binary Search Tree

AVL Tree

Red-Black Tree

Heap

Graph

LEARN

Algorithms

Complexity

Binary Search Tree

Insert

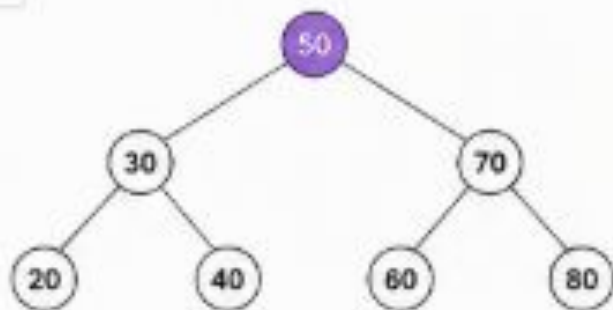
Delete

Search

Clear

Randomize

Insert: 50



Output

Inserted 50

Prev

Next

Step 2 of 2



2. Binary Search Tree Visualizer

Binary Search Tree Visualizer

Operations

Insert

Search

Delete

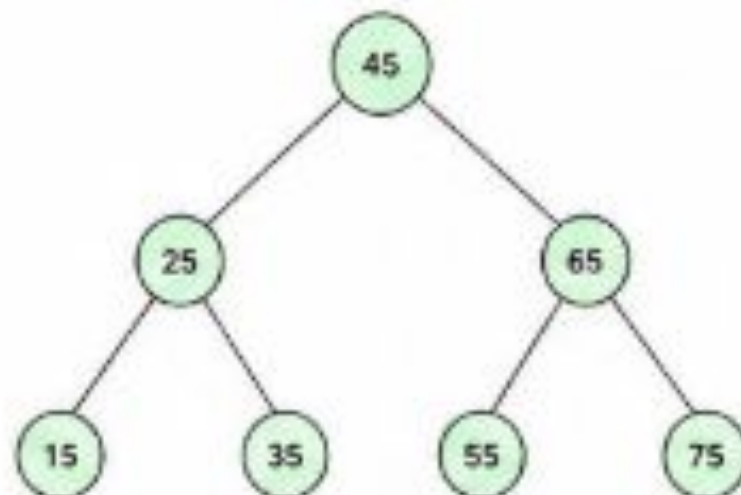
Clear

Random Tree

Input Value

45

Go



Traversal Orders

Inorder (Sorted)

15, 25, 35, 45, 55, 65, 75

Preorder

45, 25, 15, 35, 65, 55, 75

Postorder

15, 35, 25, 55, 75, 65, 45

Level Order

45, 25, 65, 15, 35, 55, 75

Insert, search and delete nodes to see the tree change in real time!



3. Binary Tree Visualizer and Converter

Input (Level Order)

1,2,3,4,5,6,7

Example: 1,2,3,4,5,6,7

Generate Tree

Clear

Traversals

Preorder

1, 2, 4, 5, 3, 6, 7

Inorder

4, 2, 5, 1, 6, 3, 7

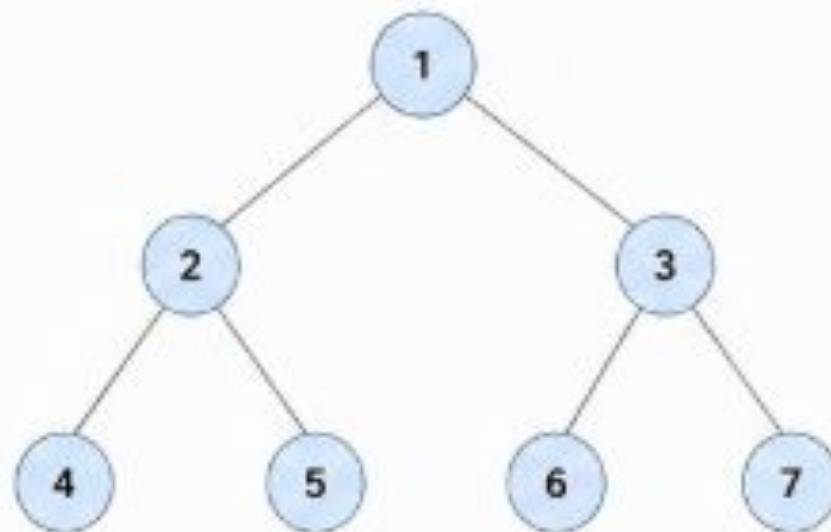
Postorder

4, 5, 2, 6, 7, 3, 1

Level Order

1, 2, 3, 4, 5, 6, 7

Generated Binary Tree



Visualize & Convert between Traversal Orders

Auto-Generation Tools

For unlimited practice challenges — fresh, algorithmic mock questions

4

LeetCode Tree Visualizer

Paste a LeetCode-serialized tree string (e.g. `[1, null, 2, 3]`) to instantly format a clean visual layout — ideal for sketching out custom traversal paths by hand.

5

Virtual Labs (IIT Hyderabad)

A government-supported interactive lab simulator that quizzes you by making you click the nodes of a randomized tree in the correct traversal sequence.



4. Virtual Labs IIITH – Tree Traversal Practice

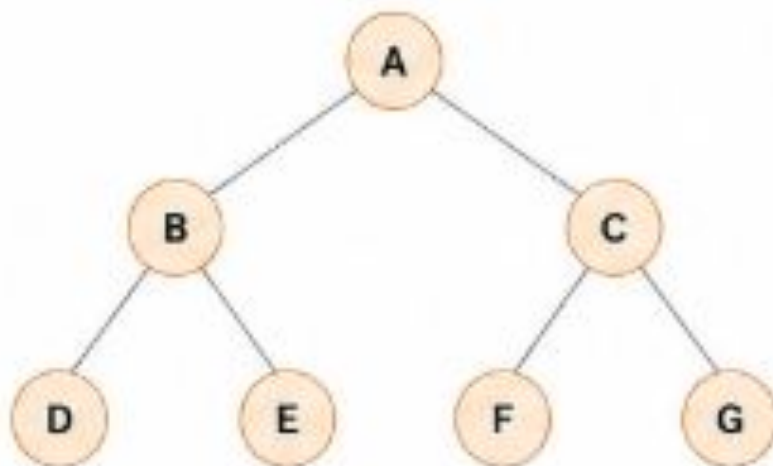
Tree Traversal Practice

Select Traversal

- ☒ Preorder
☐ Inorder
☐ Postorder
☐ Level Order

Start

Reset



Visit Order

1.
2.
4.
5.
6.
7.

Click on the nodes in the correct order for Preorder Traversal

(Root -> Left -> Right)



5. LeetCode Tree Visualize (LeetCode Tree Visualizer)



LeetCode Tree Visualizer helps you build, visualize and understand binary trees the same way they are used in LeetCode problems.



Build Trees

Create tree from array or custom input



Visualize Instantly

See the tree structure clearly and interactively



Use in Problems

Perfect for solving LeetCode tree problems

Input (Level Order) ?

[3,9,20,null,null,15,7]

Example: [1,2,3,null,4,5]

Visualize Tree

Clear

Tree Representations

Preorder (Root, Left, Right)

3, 9, 20, 15, 7

Inorder (Left, Root, Right)

9, 3, 15, 20, 7

Postorder (Left, Right, Root)

9, 15, 7, 20, 3

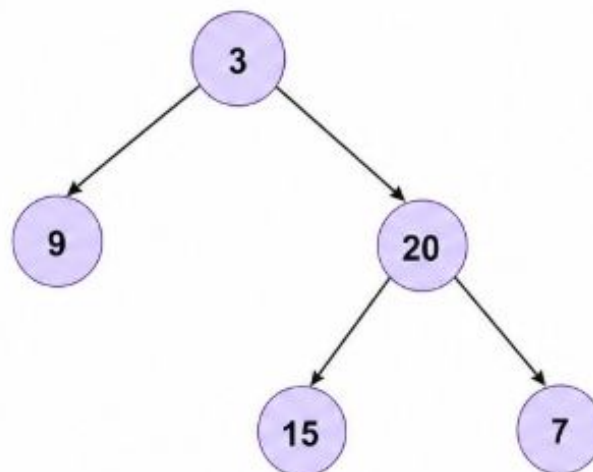
Level Order

3, 9, 20, 15, 7

Tree Visualization

☒ Show Nulls

Reset Zoom



How it works

1. Enter the tree in level order (use null for empty nodes)
2. Click "Visualize Tree" to generate the tree
3. Use the traversal outputs to understand different orders

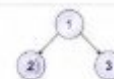
About

This visualizer helps you understand binary tree structures used in LeetCode problems.

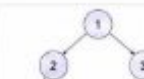
- Supports level order input
- Displays tree visually
- Shows all major traversals
- Useful for debugging & learning

Example Trees

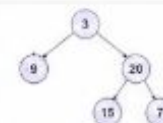
Select Example



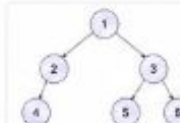
[1,2,3]



[1,2,3,null,4]



[3,9,20,null,null,15,7]



[1,2,3,4,5,6]